

Tri Le

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EDUCATION

McMaster University

Hamilton, ON

Bachelor of Software Engineering Co-op | **GPA: 3.5/4.0**

Graduation: 2027

- **Relevant Coursework:** Object-Oriented Programming, Data Structure and Algorithm, Software Development and Design Patterns, Digital Systems and Interfacing, Concurrent System
- **Awards and Achievement:** Engineering International Scholar Award (**\$10000**), Faculty of Engineering Entrance Award of Excellence (**\$3000**), PEISAA Student Athletic

EXPERIENCE

Firmware Test Engineer Intern

May 2025 - Sep 2025

QEMS Inc.

Charlotte, NC

- Tested and verified firmware on **500+** industrial control boards, ensuring **100%** functional accuracy.
- Collaborated with electrical engineers to redesign **100+** control plans to align with diverse customer specification.
- Documented test results, anomalies and failure logs, improving turnaround efficiency between QA and production by **40%**.
- Supported end-of-line testing and quality verification to maintain zero defective units shipped across multiple product lines in compliance with **ISO 9001** standard.

Open-Source Contributor

Oct 2025 - Present

Oppia - Google Summer of Code Organization

Remote

- Contributed to **Oppia**, a large-scale open-source learning platform with **900,000+** users.
- Resolve system-level issues across Oppia's **AngularJS/Angular hybrid web architecture** and **Python** backend, improving reliability and user experience.
- Collaborate through Oppia's CI/CD pipeline (**Prettier, Black, ESLint, CircleCI**) to ensure maintainable code across desktop and mobile environments.

PROJECTS

Nutrilens | **DeltaHacks - Canada's Largest Student-Run Hackathon for Change** | *React.js, Flask, OpenCV*

- Built an **AI-powered web app** that identifies meals through image recognition and provides real-time nutrition data including calories, macronutrients, and allergens.
- Engineered a full-stack solution using **React.js, Flask (Python), TensorFlow, and OpenCV**, achieving **90%+ classification accuracy** across **50+** food types.
- Integrated the **Nutritionix API** for live data retrieval and implemented responsive UI updates for user feedback.

Rescue Mission - **Path Finding and Scanning Algorithm** | *Java, JSON, Maven*

- Developed a **Java-based drone exploration algorithm** that autonomously scans and maps unknown terrain using simulated sensor feedback.
- Implemented optimized **pathfinding and fog-of-war visualization**, improving coverage efficiency by **25%**.

PingScope - **Network Performance Analyzer** | *TypeScript, WebSocket, Node.js, ICMP, Chart.js*

- Developed a real-time dashboard to monitor **latency, DNS speed, and packet loss**, processing over **1,000+ live pings per session**.
- Created a **React + TypeScript** dashboard with dynamic charts to locate unstable network nodes within **2 s**.
- Built an **ICMP- and WebSocket-based backend** for continuous data streaming, achieving **sub-100ms update latency** and improving network analysis responsiveness by **45%**.

SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS, C/C++, Verilog
- **Developer Tools:** Git, Bash, Linux, Docker, Kubernetes, Agile Methodology.
- **Libraries/Frameworks:** React.js, Node.js, Next.js, Angular, PostgreSQL, Flask, JSON